HOUSE PRICE PREDICTION PROJECT REPORT

1. Introduction

House Price Prediction is a Machine Learning project that predicts the price of a house based on various features such as house quality, living area, garage capacity, and number of bathrooms.

2. Dataset Description

Dataset Name: House Prices Dataset

Source: Kaggle

Number of Records: 1460

Number of Features: 81

3. Data Cleaning

- Checked missing values

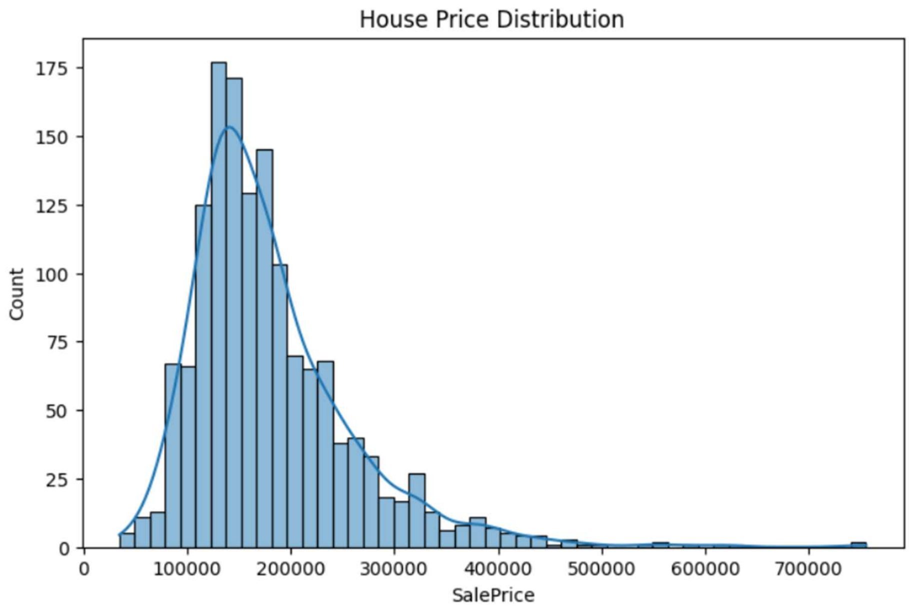
- Removed unnecessary data

- Prepared dataset for training

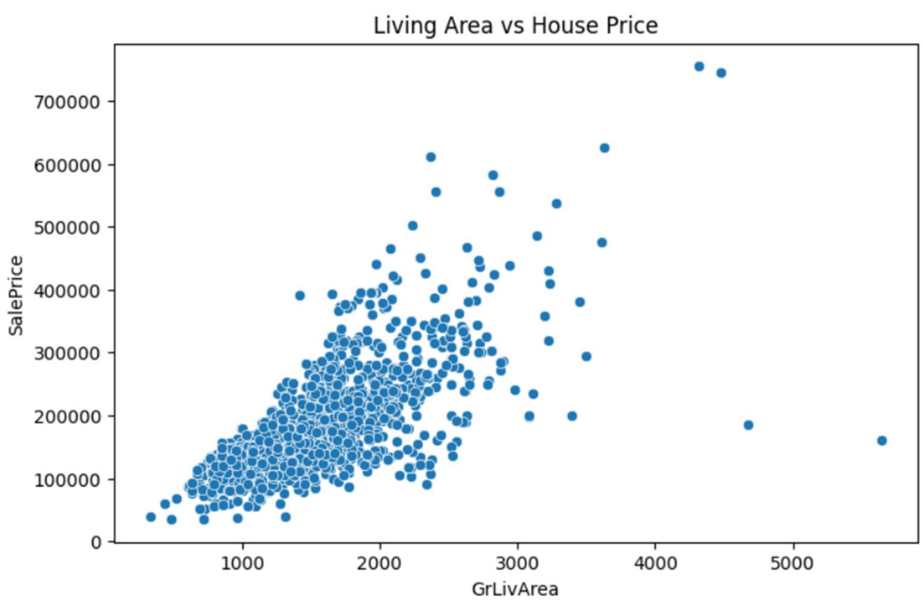
4. Data Visualization

The following visualizations were created:

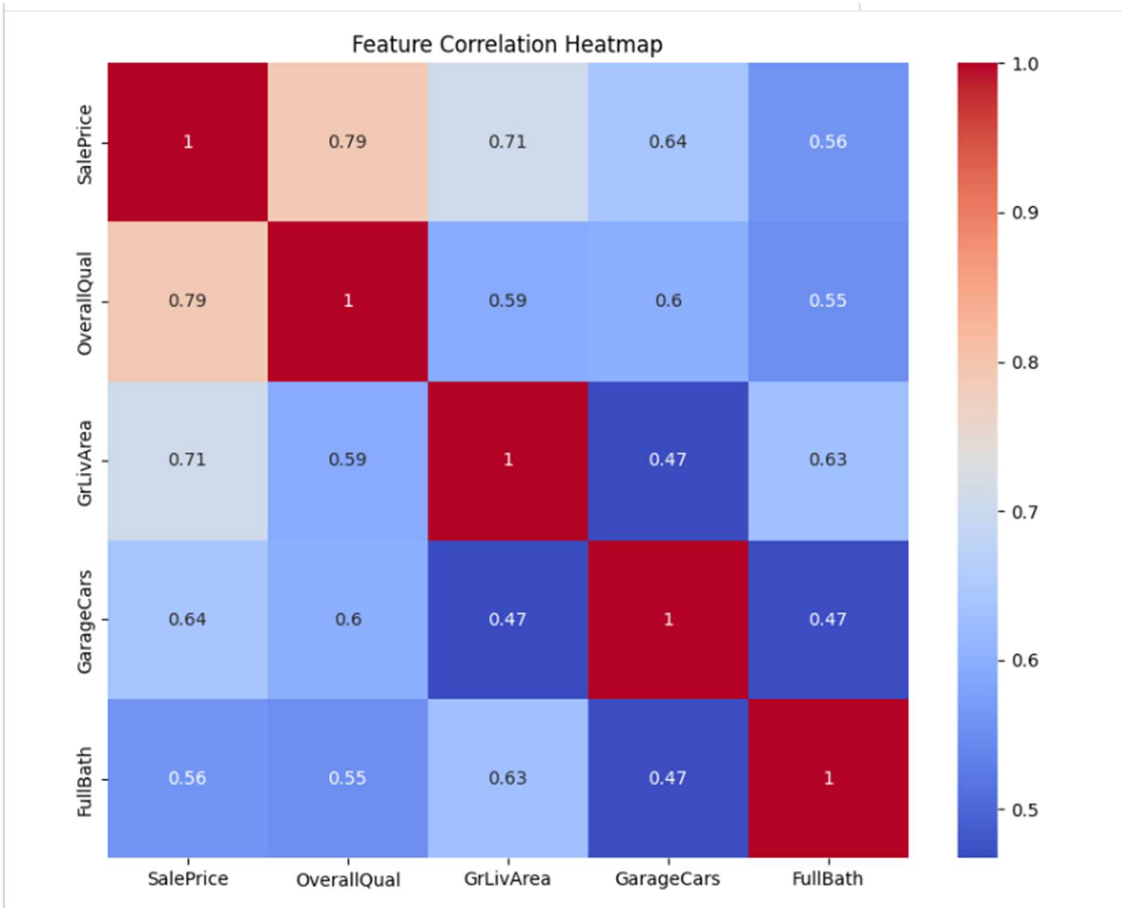
- House Price Distribution



- Living Area vs House Price



- Correlation Heatmap



5. Model Building

Algorithm Used:

Random Forest Regressor

Features Used:

- OverallQual

- GrLivArea

- GarageCars

- FullBath

Target Variable:

- SalePrice

6. Prediction Results

The model successfully predicted house prices.

Example Prediction:

House Quality: 8

Living Area: 2000 sq.ft

Garage Capacity: 2

Bathrooms: 3

Predicted Price: $282469

Category: Mid-Range Home

7. Conclusion

The House Price Prediction model successfully estimates house prices using machine learning techniques. The project demonstrates data analysis, visualization, model training, and prediction.